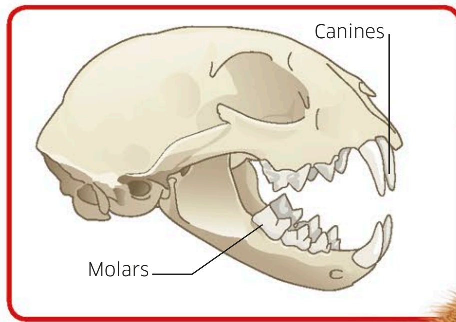
23 ft (7 m) long—the **length of an adult human intestine**. It can take **half a day for food to pass along** its length.

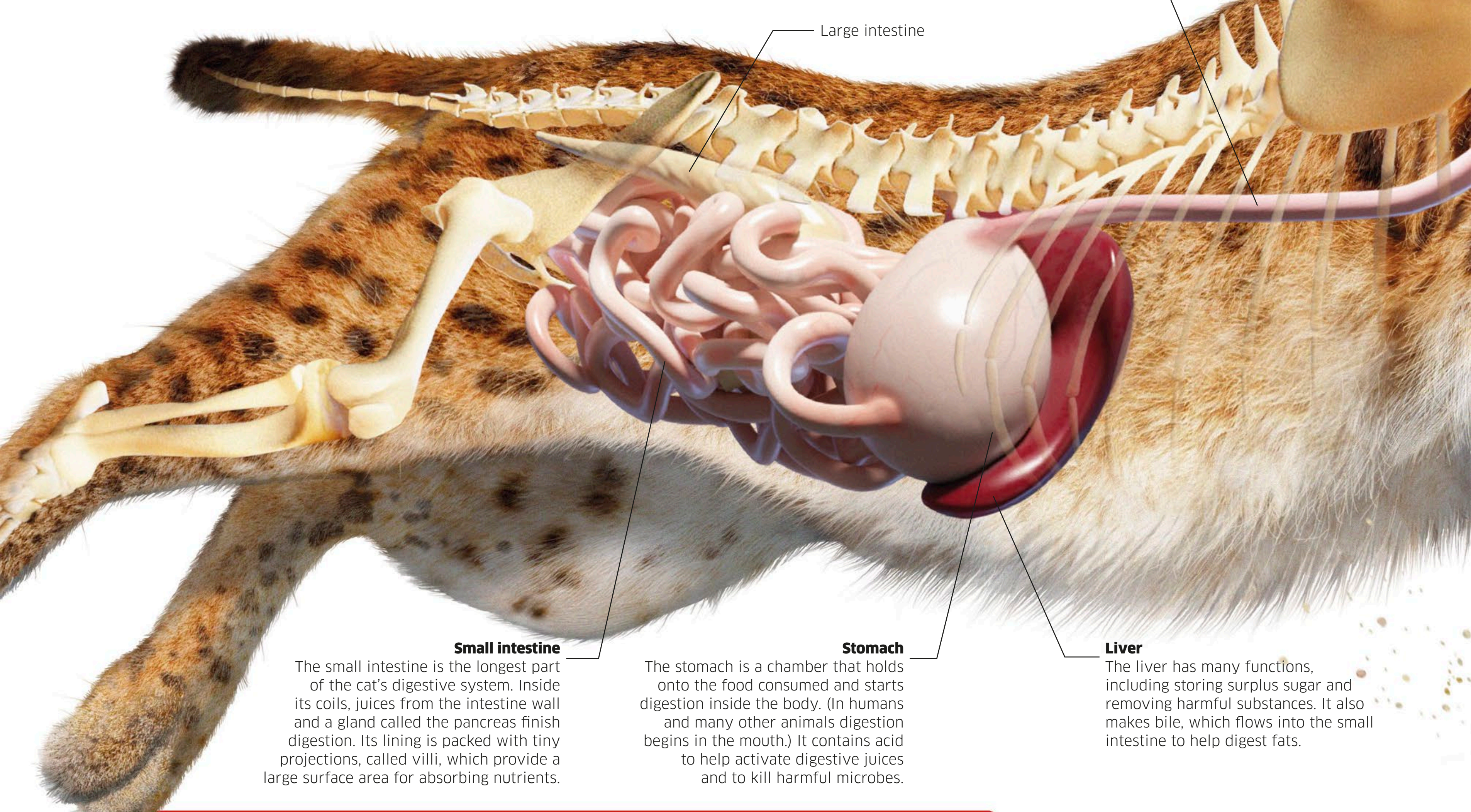
Processing food

Eating is only part of the story of how the body gets nourishment. An animal's digestive system must then break down the food so that nutrients can reach cells.

Food contains vital ingredients called nutrients, such as sugars and vitamins. Most animals eat solid food, and the digestive system has to liquefy this food inside the body so these nutrients can seep into the bloodstream. Once dissolved in the blood, they are circulated around the body to get to where they are needed—inside cells.

Carnivore teeth
Stabbing canines and sharp-edged, bone-crunching molars help the bobcat kill prey and bite through its skin and bones.





Esophagus
The esophagus (food) pipe carries lumps of swallowed food down to the stomach.

Small intestine
The small intestine is the longest part of the cat's digestive system. Inside its coils, juices from the intestine wall and a gland called the pancreas finish digestion. Its lining is packed with tiny projections, called villi, which provide a large surface area for absorbing nutrients.

Stomach
The stomach is a chamber that holds onto the food consumed and starts digestion inside the body. (In humans and many other animals digestion begins in the mouth.) It contains acid to help activate digestive juices and to kill harmful microbes.

Liver
The liver has many functions, including storing surplus sugar and removing harmful substances. It also makes bile, which flows into the small intestine to help digest fats.

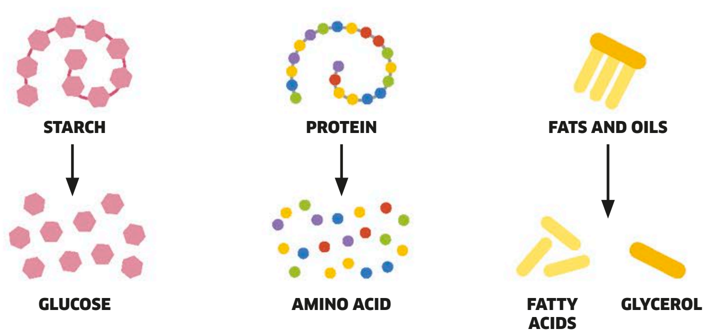
Releasing the nutrients

Biting and chewing by the mouth reduces food into manageable lumps for swallowing, but further processing is needed to extract the nutrients. Muscles in the wall of the digestive system churn food into a lumpy paste and mix it with digestive juices containing chemicals called enzymes. The enzymes help to drive chemical reactions that break big molecules into smaller ones, which are then absorbed into the blood.

Carbohydrates
Starch is digested into sugars, such as glucose.

Proteins
Proteins are digested into amino acids.

Fats
Fats and oils are broken down to release fatty acids and glycerol.



Digestive systems

A carnivorous bobcat and a herbivorous rabbit both have digestive systems filled with muscles and digestive juices to help break up their food. But they have important differences—each is adapted to the challenges of eating either chewy meat or tough vegetation.

There are more than **100 trillion bacteria** in the digestive tract.